



Worksheet W3: Instant Speed — Zooming In

"Solving the o/o Paradox"

Chapter 3 · 10 Problems · Name: _____ Date: _____

Problem 1

Question: What is the "o/o problem" when calculating instant speed?

💡 *Hint: At an instant, what are the values of distance and time?*

Problem 2

Question: If a car travels 100 meters in 5 seconds, what is its **average speed**?

💡 *Hint: Average speed = Total distance ÷ Total time.*

Problem 3

Question: What is the **limit** trick in calculus used for?

💡 *Hint: When you can't calculate something directly, you look at what it's approaching.*

Problem 4

Question: When you zoom in on a curve, what does it look like?

💡 *Hint: Think about the "Mental Zoom" from Chapter 1.*

Problem 5

Question: For the function $d = t^2$, what is the derivative (instant speed formula)?

💡 *Hint: Speed = $2 \times t$.*

Problem 6

Question: For $d = t^2$, what is the instant speed at $t = 3$ seconds?

💡 *Hint: Speed = $2t$. Plug in $t = 3$.*

Problem 7

Question: Why can't we just calculate instant speed directly by dividing distance by time?

💡 *Hint: At an instant, both distance and time are zero.*

Problem 8

Question: What is the formula for **average speed**?

💡 *Hint: Average speed = Change in distance $\div$ Change in time.*

Problem 9

Question: What does the **derivative** measure in calculus?

💡 *Hint: It measures the instant speed — the rate of change at a single moment.*

Problem 10

Question: Complete the shrinking interval table for $d = t^2$ at $t = 2$:

Interval	1 to 3	1.5 to 2.5	1.9 to 2.1	1.99 to 2.01	1.999 to 2.001
Avg Speed	4 m/s	4 m/s	4 m/s	_____	_____

💡 Hint: As the interval shrinks, what number is the average speed approaching?